Content, Not Vocabulary: Evaluating a Style Skill for LLM Prose Against Humanizers, Detectors, and Human Essays

Anonymous ACL submission

Abstract

Advice for making language-model prose read as human usually takes the form of banned-word lists, and "humanizer" tools rewrite text until detectors pass it. We test a different approach: a prompt-level style skill that audits content and structure before vocabulary and forbids adding any fact the writer did not supply. On 16 held-out briefs in ten genres, blind panels of model judges ranked the skill's output above plain generation in 47 of 48 judgments, above a word-list prompt in 43 and above the most popular open-source humanizer in 44, and it made a third as many unsupported claims as plain generation (1.17 against 3.38 per text). A judge running on a second model gave the same order. Two later versions built from judge feedback did not improve on the released writing procedure, and personal-statement rankings changed with the other texts in the panel. Three open-source detectors passed every model-written personal essay we produced, plain or not, while flagging most real admitted college essays written before ChatGPT. In a coaching experiment, the skill's coaching mode gave more useful feedback with less invented content than plain feedback. We release the skill, every version, the evaluation data and a rater survey.

1 Introduction

Readers have learned to recognize machine-written prose. Corpus studies trace sharp rises in words such as *delve* and *underscore* after the release of ChatGPT (Kobak et al., 2025; Liang et al., 2024), and instruction-tuned models use present participial clauses at two to five times the human rate (Reinhart et al., 2025). People who use these tools heavily can identify AI-written articles with near-perfect accuracy, and they cite vocabulary, structure and lack of originality as their evidence (Russell et al., 2025).

Two responses dominate practice. The first is the banned-word list, distributed in prompts, style guides and community "skills". The second is the commercial humanizer, which rewrites text until a detector scores it as human (Roe et al., 2026). Both treat the surface as the problem. Wikipedia's editors, whose guide most word lists derive from, warn against exactly this: "Please do not merely treat these signs as the problems to be fixed; that could just make detection harder" (Wikipedia contributors, 2026). Both responses also ignore a second failure of model prose: models fill gaps with plausible details that nobody supplied (Ji et al., 2023), and in personal writing those details are feelings and memories that belong to someone else.

We built and evaluated an alternative: a style skill (a set of instructions an agent loads when a task matches) that orders its audit from content to staging, inflation, rhythm, tone and only then vocabulary, and that never adds a fact absent from the writer's material. For college application essays it switches to a coaching mode and does not write the essay. This paper reports what happened when we tested it.

Our contributions are:

- An evidence-graded design for a writing skill, built from a review of about 40 sources on AI-writing tells and of 15 existing open-source humanizer skills, and released with all seven versions.
- A controlled comparison on 20 development and 16 held-out briefs in ten genres against plain generation, a word-list prompt and a popular humanizer, scored by blind judge panels (including a judge on a second model) and by three open-source detectors.
- A detector audit against human writing collected before ChatGPT. On personal essays the detectors invert: they flag admitted essays and pass model text, plain or not.

- A record of the iterations, including two versions that did not beat their predecessor, and a measurement effect: personal-statement rankings depend on which other texts share the panel.

- A coaching experiment comparing the skill's coaching mode with plain feedback on simulated student drafts.

2 Related work

Stylistic signatures of LLM text. Excess-vocabulary analyses of PubMed and peer reviews show that LLM adoption shifted the frequency of style words, mostly verbs and adjectives (Kobak et al., 2025; Liang et al., 2024; Juzek and Ward, 2025; Geng and Trotta, 2024). Beyond words, instruction-tuned models write a noun-heavy, informationally dense register (Reinhart et al., 2025), reuse syntactic templates from pre-training data (Shaib et al., 2024), and overuse corrective framings such as "not X, but Y" even where X is not a live alternative (Peng et al., 2026). Professional writers editing model paragraphs agree on a taxonomy of idiosyncrasies led by clichés and unnecessary exposition (Chakrabarty et al., 2025), and an expert taxonomy of "slop" finds that syntax-only features predict slop judgments better than vocabulary (Shaib et al., 2026). Token-level bans on such patterns degrade output quickly (Paech et al., 2026). AI generation leaves a stable stylometric footprint that AI editing of human text does not reproduce (Shan et al., 2026).

Detecting AI text. Detectors range from trained classifiers to zero-shot statistics (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024). The RAID benchmark shows that many are nearly inoperable at low false-positive rates and fail under attacks or unseen generators (Dugan et al., 2024). Paraphrasing defeats watermark-, classifier- and zero-shot detectors (Krishna et al., 2023; Sadasivan et al., 2025), and simple prompting lowers already modest accuracy (Perkins et al., 2024; Weber-Wulff et al., 2023). Detectors misclassify non-native English writing as AI at high rates (Liang et al., 2023) and flag formal human prose (Creanga and Dinu, 2026); light, guideline-compliant AI editing is flagged more often than humanizer output (Karr et al., 2026). Untrained humans detect AI text at about chance (Clark et al., 2021; Jakesch et al., 2023), while frequent LLM users do far better (Russell et al., 2025). Text watermarks offer a provenance signal that does not depend on style (Kirchenbauer et al., 2023; Dathathri et al., 2024).

LLMs and admissions essays. LLM-written admissions essays are linguistically distinct from those of 30,000 applicants and resemble the essays of privileged applicants (Lee et al., 2025; Alvero et al., 2024); essay content and style already tracked household income before LLMs (Alvero et al., 2021). Estimated LLM use in essays rose sharply in 2024 (Lee et al., 2026), and in one graduate program, essays flagged as AI-written were associated with lower admission (Isley et al., 2026). AI writing help homogenizes writing across authors (Padmakumar and He, 2024; Doshi and Hauser, 2024; Agarwal et al., 2025) and can shift meaning even when limited to grammar (Abdulhai et al., 2026), and heavy rewriting erases authorship signals (Malik and Awan, 2026).

LLMs as judges. Strong LLM judges agree with human preferences about as often as humans agree with each other, but show position, verbosity and self-enhancement biases (Zheng et al., 2023; Wang et al., 2024), and models can recognize and prefer their own outputs (Panickssery et al., 2024). We discuss how these biases bear on our design in Section 7.

3 The skill

The released skill is a Markdown instruction file (173 lines) with four reference files: a catalog of tells with fixes, a personal-essay guide, academic-register rules, and before/after examples. An agent loads it when a writing task matches its description. We built it from two reviews: a survey of 15 open-source humanizer skills, which supplied the procedure of Chen (2026) and guardrails from two other MIT-licensed skills, and an evidence-graded catalog of tells assembled from about 40 sources.

Audit order. The catalog showed that the strongest measured signals are structural (participial tails, corrective framing, importance flags, uniform sentence length) and that expert readers rely on content as much as words (Russell et al., 2025; Shaib et al., 2026). The skill therefore audits a draft in a fixed order: content (the "portability test": could this sentence move unchanged into someone else's text?), staging, inflation, stated lessons, rhythm, paragraph endings, tone, vocabulary, and

format. A strong tell justifies an edit on one sighting; weak tells, such as a single transition word, only when several cluster.

Hard rules. The skill never adds a fact, name, number, quote, feeling or memory that the user or the source did not provide; where a sentence needs one, it asks or leaves a bracketed question. It keeps every claim and hedge when editing, uses no evasion tricks (typos, rare synonyms, look-alike characters), and makes minimal edits to human-written text.

Over-correction checks. Humanizer rewrites create tells of their own: the humanizer's own blind test reports more uniform rhythm and more aphoristic closers after rewriting (Chen, 2026). The final check lists the last sentence of every paragraph and rewrites when more than two are short or quotable, and checks that a personal essay still has an arc and a line of reflection.

Modes. *Write* runs the full procedure; for personal essays it drafts in two passes, first for shape and then as an editor of its own draft (Section 5.4). *Edit* applies the audit to the user's own draft with minimal changes. *Rebuild*, for drafts the user says an AI wrote, first checks every fact, quote and reading against the user's material; if the draft was built from that material it keeps the checked evidence and writes fresh sentences, and if it was built from the topic alone it treats the draft's examples and plan as the stock answer and writes from the first step. *Detect* quotes passages and names patterns without guessing authorship. *Coach*, used for application essays, interviews the student, proposes a structure, critiques drafts and lists corrections, and does not write the essay or example sentences built from the student's story.

4 Evaluation

We evaluated the skill in three rounds: a development round on 20 briefs, a version comparison on the same briefs, and a held-out round on 16 new briefs written after the final version was fixed. A separate experiment tests the coaching mode.

Briefs. The development set has 20 briefs in five genres, four each (Table 3). Four fictional students, each with about 1,100 to 1,600 words of interview notes and answers to the skill's depth questions, supply the material for a Common App personal statement and a 250-word supplemental answer. The other briefs are 11th-grade argument essays and science-club explainers with no facts file, and literary analyses restricted to four passages from a public-domain novel or play, copied from Project Gutenberg and checked by script. The held-out set has 16 briefs: two new students (a personal statement and a short answer each), two arguments, two explainers, two literary analyses, and six genres the skill was never tested on: an email to a teacher, a cold email to a professor, a cover letter, a team blog post, the discussion section of a lab report, and a school news article. Every brief names the only permitted source of specifics; the people in the fact files are fictional.

Conditions. Fresh [generator model, name and version] agents wrote every text. **Plain** (P): the model's default writing. **Word list** (L): the same request plus an instruction to avoid 30 common AI words and constructions and to vary sentence length. **Humanizer** (H): the most popular open-source humanizer skill (Chen, 2026), applied to P's draft. The skill conditions were the Write mode, the Edit mode applied to P's draft, and from version 5 on a Rebuild mode for drafts the user says an AI wrote. The development round compared P, L, H, Edit and Write (version 4); the version comparison compared P, L, version 4 Write, version 5 Write and version 5 Rebuild; the held-out round compared P, L, H, version 4 Write, version 6 Write and version 6 Rebuild.

Judges. Judge agents read all texts for a brief in a shuffled order without labels: a strict editor, and the genre's real reader (an admissions officer, an English or science teacher, the recipient of the email, a hiring manager, a team mentor, a newspaper adviser). Each ranked the texts, scored each from 1 to 10 on quality, human-likeness, specificity, voice and structure, rated AI suspicion from 1 to 5, and quoted every claim not supported by the facts file (or every invented statistic, source or factual error when there was none). In the held-out round a third judge, a second genre reader, ran on a different model ([second judge model, name and version]). We report mean rank with bootstrap confidence intervals, a Friedman test across conditions on per-brief mean ranks, and pairwise Wilcoxon signed-rank tests.

Detectors. We scored every text with three open-source classifiers: a DeBERTa-v3-large model trained on RAID (Desklib, 2025), a RoBERTa-base

model (Fakespot, 2025), and a RoBERTa-large model whose training data includes Anthropic-model output (SuperAnnotate, 2024). Texts are split into chunks of about 350 words, and document scores are length-weighted means. We report the share below 20%, the threshold below which Turnitin hides its AI score.

Human baselines. To see how the detectors treat human writing, we scored three sets collected before ChatGPT: 16 admitted college essays published by Hamilton College (classes of 2018 and 2022) and Connecticut College (class of 2025); 18 top-scored argument essays from the PERSUADE 2.0 corpus (Crossley et al., 2024); and three science explainers published in 2020.

Coaching experiment. For each of the six fictional students, an agent playing the student wrote a first draft of the personal statement from their notes, with no style guidance. The draft then received feedback from the skill's Coach mode or from the plain model asked for feedback, and the student agent revised it with each feedback separately. Two blind judges ranked the first draft and both revisions, and a counselor agent rated both pieces of feedback for usefulness and counted the essay-ready sentences each wrote for the student. We also measured how much of each revision was copied from the feedback it received: the share of the revision's word 6-grams that occur in the feedback but in neither the first draft nor the student's notes.

5 Results

5.1 Held-out briefs

Table 1 gives the main result. On 16 briefs written after the final design was fixed, ten genres among them, the skill's Write mode (version 4) ranked first in 23 of 48 judgments and had a mean rank of 1.92 out of 6, against 5.65 for plain generation (Friedman $\chi^2 = 51.0$, $p < 10^{-8}$). It ranked above plain generation in 47 of 48 judgments, above the word-list prompt in 43 and above the humanizer edit in 44 (Wilcoxon $p \leq 0.003$ for each). It made 1.17 unsupported claims per text, against 3.38 for plain generation and 2.98 for the word-list prompt. The Rebuild mode, applied to the plain draft, ranked above the humanizer applied to the same draft in 42 of 48 judgments ($p = 0.0008$) and had the fewest unsupported claims of any condition (0.79).

Version 6, which added rules derived from a diagnosis of the development judgments, did not improve on version 4: it ranked above version 4 in 21 of 48 judgments ($p = 0.35$). Its argument essays, built around a worked numerical case, were called "arithmetic-heavy for an English class." We release version 4's Write procedure with version 6's Rebuild mode added (Section 3).

Condition	Rank	1st	Human	Unsup.
Plain	5.65	0	4.5	3.38
Word list	4.17	3	6.5	2.98
Humanizer edit	4.25	2	5.7	1.88
Skill write (v4)	**1.92**	**23**	**7.2**	1.17
Skill write (v6)	2.42	17	7.0	0.96
Skill rebuild (v6)	2.60	3	7.1	**0.79**

Table 1: Held-out evaluation: 16 briefs, three blind judges each (48 judgments). Rank: mean rank among six texts (1 = best); 1st: first places; Human: judged human-likeness (1–10); Unsup.: claims per text that judges found unsupported by the facts file or wrong.

The third judge, which ran on a different model, gave the same order (version 4 1.94, version 6 2.25, Rebuild 2.44, word list 4.19, humanizer 4.81, plain 5.38), so the ranking does not depend on one judge model, within one model family.

By genre, version 4 ranked best on personal statements (1.33), arguments (1.33), literary analysis (1.33) and the news article (1.00). It ranked worst among the skill conditions on the blog post (4.67), where the word-list prompt won (1.67), and the word-list prompt also edged it on the cover letter (2.00 against 2.33). With one or two briefs per genre these are descriptive, but both losses are in casual first-person genres, where the skill's audit may strip the looseness the genre wants.

5.2 Development round and the version comparison

On the 20 development briefs, version 4's Write mode ranked first in 28 of 40 judgments (mean rank 1.45 of 5; plain 4.62; word list 2.27; humanizer 3.48; Edit mode on the plain draft 3.17). Version 5 added a plain student voice for school essays, a Rebuild mode and a personal-essay checklist. In a new five-way panel on the same briefs it raised judged human-likeness (7.45 against 6.90) but ranked below version 4 in 28 of 40 judgments: judges caught simplified reasoning, and the example stance line in the rule ("I think the policy is wrong") became the opening of three of four argument essays.

The version comparison also exposed a measurement problem. The plain and version 4 texts were

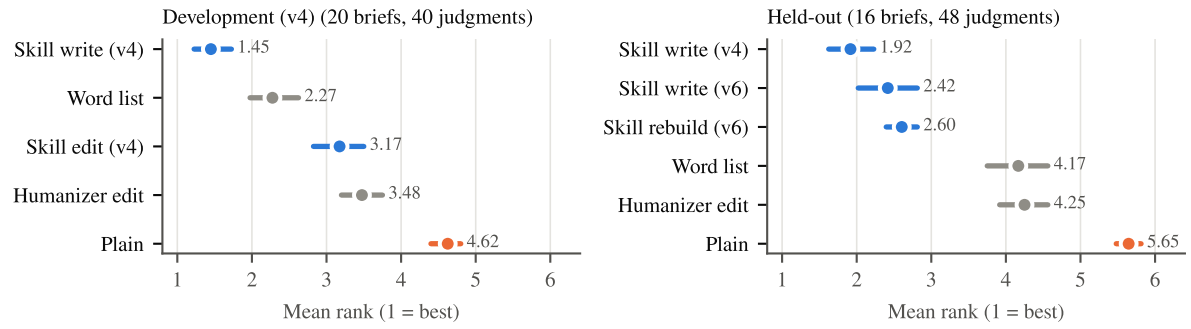


Figure 1: Mean rank with 95% bootstrap intervals (1 = best). Left: development briefs, five texts per panel. Right: held-out briefs, six texts per panel, three judges each. Blue: the skill; orange: plain generation; grey: other approaches.

identical in both panels, yet on personal statements version 4 ranked above plain in 8 of 8 judgments in the first panel and in only 3 of 8 in the second. The first panel contained two edits of the plain draft, and judges noticed the shared sentences and marked the whole family down. In the other genres the comparison held (8 of 8 in both panels). Rankings of personal statements depend on what they are ranked against, which is one reason we report the held-out round as the main result.

5.3 Detectors and human writing

Figure 2 shows the held-out scores. All 16 texts written with the skill and all 16 rebuilds scored below 20% on all three detectors, against 9 of 16 for plain generation and 12 each for the word-list prompt and the humanizer edit. The development round agreed (17 of 20 skill texts against 9 of 20 plain). The advantage came from the school and technical genres: on personal statements and short answers every condition passed, plain generation included, with no score above 3% in either round. The human baselines went the other way. None of the 16 admitted college essays scored below 20% on the third detector, whose mean for them was 61%, and only 5 did on the first two; 2 of the 18 PERSUADE essays passed all three. On personal essays these detectors passed every model-written text we produced and flagged most human ones.

5.4 Personal statements: arc against fidelity

Personal statements were the hardest genre throughout development (Table 2). Plain generation built a better arc by writing connective feelings and causes the student never stated (3 to 11 unsupported claims per essay), and judges preferred it. Forbidding invention flattened the skill's first three versions; ver-

Version	Change	vs. plain	Unsup.
v1	audit while drafting	0–4	–
v2	stricter sample mode	0–4	–
v3	countable checks	0–4	1 vs 16
v4	two-pass drafting	1–3	6 vs 19

Table 2: Personal-statement iterations in development (pairwise judgments of better writing against plain generation, depth notes for v3–v4; unsupported claims summed over the same pairs). Small samples: 2–4 pairs per cell.

sion 3, for example, "walks through the notes in order; climax in paragraph 7 of 9." Two-pass drafting in version 4 (write for shape first, then audit the draft as an editor) recovered part of the arc: its first win against plain generation on the original notes, with 4 unsupported claims against 14. Giving both conditions the student's answers to depth questions raised everyone's quality and halved plain generation's invention. On the two held-out students, version 4 ranked first in 5 of 6 personal-statement judgments with 0.67 unsupported claims per essay, against 4.83 for plain generation.

5.5 Coaching

Both kinds of feedback improved the first drafts: the draft ranked last in all 12 judgments. The revision made with Coach-mode feedback ranked above the one made with plain feedback in 7 of 12 judgments (not significant), with fewer unsupported claims (1.67 against 2.75) and lower AI suspicion (1.83 against 2.08). The clearer difference was in the feedback itself. The counselor judge preferred Coach mode's feedback for all six students (usefulness 8.67 against 7.17 of 10) and found less invented content in it (1.17 items per student against 5.17). Plain feedback rarely wrote

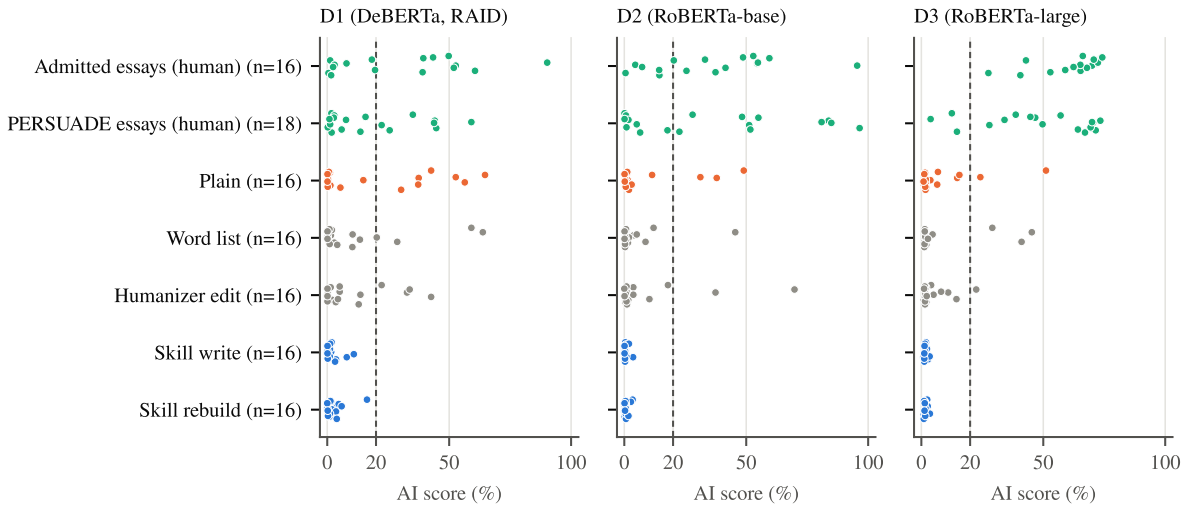


Figure 2: Detector scores per text on the held-out briefs, with human writing collected before ChatGPT for comparison. The dashed line is 20%. D3's scores saturate near 73%.

whole sentences for the student (1.33 per response), but it supplied the essay's interpretation: one response handed a student "the essay's interpretive center" that he never said, which is the substantive content that admissions policies forbid. The simulated students copied almost nothing from either feedback (under 1% of word 6-grams), so this experiment cannot measure what a rushed real student would copy. Coach mode's responses were also about 1.5 times longer (2,264 words against 1,468), which may be more than a student will read.

6 Discussion

Why content first. Judges' reasons point to the same mechanism across genres. The texts they ranked first argued from one specific organizing idea, used the writer's own facts and kept them straight, and ended on content; the texts they ranked last had the stock plan of the genre (a three-reason thesis, a sensory hook, a summary close). The word-list prompt changed the vocabulary and the voice, and judges rated it more human than plain generation, but it kept the stock plan and invented almost as much (2.98 unsupported claims per text against 3.38). The humanizer edit removed tells sentence by sentence and kept the draft's plan, and judges recognized edits of the same draft as one family. The skill's gains came from what the vocabulary pass cannot reach: what the text claims, in what order, and from which facts.

Iterating on judges. Diagnosing judge comments produced plausible rules that did not transfer. Version 5's student-voice rule raised human-likeness and lowered reasoning quality, and its example line became a template; version 6's argument rules, distilled from 16 judgments that praised one move, produced essays judges called arithmetic-heavy on new topics. Both regressions were caught only because each version was compared with its predecessor on the same panel and, finally, on held-out briefs. We think style skills, like models, need a held-out set: the development judges' taste is part of what gets fitted.

Detectors in admissions. On personal essays the three detectors got the direction wrong: they scored real admitted essays high and model essays written with the skill low. A plausible reason is domain. Two of the detectors were trained on RAID, whose eight domains (scientific abstracts, book summaries, news, poetry, recipes, Reddit posts, reviews and Wikipedia) do not include the personal essay (Dugan et al., 2024). In most of those domains human writing is uneven and specific and model writing is polished and general. Admitted essays are polished, edited many times, and full of the reflective phrasing that admissions advice recommends, so they look like the model side of the training data; a model told to use concrete, uneven detail looks like the human side. Whatever the cause, a detector score is not evidence that an applicant used AI. Colleges that forbid substantive AI content (Common App, 2023; Brown University Office of College Admission, 2026; Yale Undergraduate Admissions, 2026) have to enforce it

some other way, and our results suggest that style-based detection would fall hardest on applicants who write in the polished register that essay coaching teaches.

Coaching versus ghostwriting. Personal statements were the one genre where the skill's writing did not clearly win, and the reason is instructive. The essays that judges liked most built an arc, a turning point and a closing insight, and plain generation built them by inventing feelings and causes the notes did not contain. When the skill forbade that, it lost some of the arc. The material that makes an application essay work (what the student actually thought, what they still have not resolved) exists only in the student's head, so the honest way for a model to help is to ask for it. Coach mode does this and leaves the writing to the student, which also keeps the essay inside policies that allow brainstorming and proofreading but not AI-written content (Yale Undergraduate Admissions, 2026).

7 Conclusion

A style skill that fixes content and structure before vocabulary, and that never adds facts, produced writing that blind model judges preferred to plain generation, a word-list prompt and a popular humanizer on briefs it was not developed on, with a third of plain generation's unsupported claims. Its coaching mode gave more useful feedback than plain feedback while leaving the writing to the student. The detectors we tested could not tell its personal essays, or plain model essays, from human ones, and flagged most real admitted essays, so they should not be used as evidence in admissions. The open questions are whether human readers agree with the model judges, which our released survey is built to test, and how to keep casual first-person genres loose while auditing them.

Limitations

Model judges. All judges are language models from the same family as the generator, and models can recognize and favor their own output (Panickssery et al., 2024). Self-preference does not by itself favor one of our conditions, since every condition was produced by the same model, but judges may share the generator's taste for a particular kind of polish. We shuffled labels, used two reader lenses, and in the held-out study added a judge running on a different model tier; we did not test judges from other model families or recruit enough human raters. A human rating study with the released survey is the most important missing piece.

Test material. The student personas are fictional and were written by a model; real interview notes are shorter, messier and less quotable. Coach mode was checked on one real student draft and one pressure test, which shows it can work but not how often it does.

Detectors. We used three open-source classifiers. Commercial detectors used by schools (Turnitin, GPTZero) were not tested, so the pass rates here do not predict their scores. The human baselines are small (16 admitted essays, 18 PERSUADE essays) and the admitted essays were selected and edited for publication by the colleges.

Scope and development. One generator model wrote every text, in English, for genres common in US high schools. Version 5 of the skill was revised after the phase-2 results on the same 20 briefs, so its phase-2 scores are in-sample; the held-out study uses 16 new briefs, but they were written by the same team that built the skill. Per-genre results rest on two to four briefs each and should be read as descriptive.

Ethical considerations

Dual use. The skill lowers detector scores as a side effect of better writing, and a student could use it to pass off model-written work as their own. We took three steps. The skill uses no evasion tricks (typos, homoglyphs, invisible characters), so it adds nothing to the attack literature on detectors (Krishna et al., 2023; Sadasivan et al., 2025). For application essays it coaches and does not write, and it states institutional AI policies plainly. And it tells users that recent model output may carry a provenance watermark (BleepingComputer, 2026): style editing does not make model text the user's own work.

Detector developers. Because we report low detector scores for model text and high scores for human essays, we [notified the developers of all three detectors on <date>], at least 30 days before any preprint.

Human writing and raters. We scored published admitted essays and PERSUADE essays and report only their scores; we do not redistribute their

texts. No real applicant's essay appears in the released data. Any human rating study uses volunteers who give informed consent, collects no names or contact details, and [had approval from <school or teacher>].

AI use statement

AI agents ([generator model, name and version], Anthropic) did a large share of this work: they surveyed the literature and existing skills, drafted every version of the skill, generated all evaluation texts, acted as judges, wrote the analysis scripts, and drafted this paper. The development log, prompts, scripts and every intermediate version are released. [The author describes here, accurately, what they did themselves: e.g. which decisions they made, which numbers and citations they checked against the data and sources, and which sections they rewrote.]

References

Marwa Abdulhai, Isadora White, Yanming Wan, Ibrahim Qureshi, Joel Z. Leibo, Max Kleiman-Weiner, and Natasha Jaques. 2026. How LLMs distort our written language. *arXiv preprint arXiv:2603.18161*.

Dhruv Agarwal, Mor Naaman, and Aditya Vashistha. 2025. AI suggestions homogenize writing toward Western styles and diminish cultural nuances. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–21. Association for Computing Machinery.

A. J. Alvero, Jinsook Lee, Alejandra Regla-Vargas, René F. Kizilcec, Thorsten Joachims, and Anthony Lising Antonio. 2024. Large language models, social demography, and hegemony: comparing authorship in human and synthetic text. *Journal of Big Data*, 11(1):138.

AJ Alvero, Sonia Giebel, Ben Gebre-Medhin, Anthony Lising Antonio, Mitchell L. Stevens, and Benjamin W. Domingue. 2021. Essay content and style are strongly related to household income and SAT scores: Evidence from 60,000 undergraduate applications. *Science Advances*, 7(42):eabi9031.

Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, and Yue Zhang. 2024. Fast-DetectGPT: Efficient zero-shot detection of machine-generated text via conditional probability curvature. In *The Twelfth International Conference on Learning Representations*.

BleepingComputer. 2026. How Anthropic plans to watermark Claude's AI-generated text. https://bleepingcomputer.com/news/artificial-intelligence/how-anthropic-plans-to-watermark-claudes-ai-generated-text. News report, accessed 20 October 2026.

Brown University Office of College Admission. 2026. Integrity in the application process. https://admission.brown.edu/apply/how-apply/integrity-application-process. Accessed 20 October 2026.

Tuhin Chakrabarty, Philippe Laban, and Chien-Sheng Wu. 2025. Can AI writing be salvaged? mitigating idiosyncrasies and improving human-AI alignment in the writing process through edits. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–33. Association for Computing Machinery.

Siqi Chen. 2026. Humanizer. GitHub repository, https://github.com/blader/humanizer. Agent skill (SKILL.md metadata version 3.1.0); MIT license, copyright 2025 Siqi Chen; accessed 2026-10-20.

Elizabeth Clark, Tal August, Sofia Serrano, Nikita Haduong, Suchin Gururangan, and Noah A. Smith. 2021. All that's 'human' is not gold: Evaluating human evaluation of generated text. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7282–7296, Online. Association for Computational Linguistics.

Common App. 2023. Fraud policy. https://www.commonapp.org/fraud-policy/. Last updated August 2023; accessed 20 October 2026.

Claudiu Creanga and Liviu Dinu. 2026. Ontological instability and statistical amplification: The paradox of "humanizing" LLM-generated text. *arXiv preprint arXiv:2610.03110*.

S. A. Crossley, Y. Tian, P. Baffour, A. Franklin, M. Benner, and U. Boser. 2024. A large-scale corpus for assessing written argumentation: PERSUADE 2.0. *Assessing Writing*, 61:100865.

Sumanth Dathathri, Abigail See, Sumedh Ghaisas, Po-Sen Huang, Rob McAdam, Johannes Welbl, Vandana Bachani, Alex Kaskasoli, Robert Stanforth, Tatiana Matejovicova, Jamie Hayes, Nidhi Vyas, Majd Al Merey, Jonah Brown-Cohen, Rudy Bunel, Borja Balle, Taylan Cemgil, Zahra Ahmed, Kitty Stacpoole, and 5 others. 2024. Scalable watermarking for identifying large language model outputs. *Nature*, 634(8035):818–823.

Desklib. 2025. desklib/ai-text-detector-v1.01. Hugging Face model card, https://huggingface.co/desklib/ai-text-detector-v1.01. DeBERTa-v3-large fine-tuned on RAID; MIT license; last updated 17 Feb 2025; accessed 2026-10-20.

Anil R. Doshi and Oliver P. Hauser. 2024. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28):eadn5290.

Liam Dugan, Alyssa Hwang, Filip Trhlík, Andrew Zhu, Josh Magnus Ludan, Hainiu Xu, Daphne Ippolito, and Chris Callison-Burch. 2024. RAID: A shared benchmark for robust evaluation of machine-generated text detectors. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12463–12492, Bangkok, Thailand. Association for Computational Linguistics.

Fakespot. 2025. fakespot-ai/roberta-base-ai-text-detection-v1. Hugging Face model card, `https://huggingface.co/fakespot-ai/roberta-base-ai-text-detection-v1`. RoBERTa-base fine-tune; Apache-2.0 license; last updated 25 Feb 2025; accessed 2026-10-20.

Mingmeng Geng and Roberto Trotta. 2024. Is ChatGPT transforming academics' writing style? *arXiv preprint arXiv:2404.08627*.

Abhimanyu Hans, Avi Schwarzschild, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, and Tom Goldstein. 2024. Spotting LLMs with Binoculars: Zero-shot detection of machine-generated text. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 17519–17537. PMLR.

Calvin Isley, Johann D. Gaebler, and Sharad Goel. 2026. AI-written admissions essays are widespread but penalized. *arXiv preprint arXiv:2609.22549*.

Maurice Jakesch, Jeffrey T. Hancock, and Mor Naaman. 2023. Human heuristics for AI-generated language are flawed. *Proceedings of the National Academy of Sciences*, 120(11):e2208839120.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Tom S. Juzek and Zina B. Ward. 2025. Why does ChatGPT "delve" so much? exploring the sources of lexical overrepresentation in large language models. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 6397–6411, Abu Dhabi, UAE. Association for Computational Linguistics.

Jonathan A. Karr, Grigorii Khvatskii, Ting Hua, and Nitesh V. Chawla. 2026. Why AI detection fails for academic integrity. *arXiv preprint arXiv:2608.11256*. ArXiv comment: accepted to ACM AI Leadership Summit.

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. 2023. A watermark for large language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 17061–17084. PMLR.

Dmitry Kobak, Rita González-Márquez, Emőke-Ágnes Horvát, and Jan Lause. 2025. Delving into LLM-assisted writing in biomedical publications through excess vocabulary. *Science Advances*, 11(27):eadt3813.

Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. 2023. Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. In *Advances in Neural Information Processing Systems*, volume 36, pages 27469–27500.

Jinsook Lee, AJ Alvero, Thorsten Joachims, and René Kizilcec. 2025. Poor alignment and steerability of large language models: Evidence from college admission essays. *arXiv preprint arXiv:2503.20062*.

Jinsook Lee, Conrad Borchers, AJ Alvero, Thorsten Joachims, and Rene F. Kizilcec. 2026. The digital divide in generative AI: Evidence from large language model use in college admissions essays. In *Proceedings of the Thirteenth ACM Conference on Learning @ Scale*, pages 179–190. Association for Computing Machinery.

Weixin Liang, Zachary Izzo, Yaohui Zhang, Haley Lepp, Hancheng Cao, Xuandong Zhao, Lingjiao Chen, Haotian Ye, Sheng Liu, Zhi Huang, Daniel A. McFarland, and James Y. Zou. 2024. Monitoring AI-modified content at scale: A case study on the impact of ChatGPT on AI conference peer reviews. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 29575–29620. PMLR.

Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou. 2023. GPT detectors are biased against non-native English writers. *Patterns*, 4(7):100779.

Ushna Malik and Moiz Sadiq Awan. 2026. The assistant erased you: Measuring loss of authorship signals in AI-mediated communication. *arXiv preprint arXiv:2608.00926*.

Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D. Manning, and Chelsea Finn. 2023. DetectGPT: Zero-shot machine-generated text detection using probability curvature. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 24950–24962. PMLR.

Vishakh Padmakumar and He He. 2024. Does writing with language models reduce content diversity? In *The Twelfth International Conference on Learning Representations*.

Samuel J. Paech, Allen G. Roush, Judah Goldfeder, and Ravid Shwartz-Ziv. 2026. Antislop: A comprehensive framework for identifying and eliminating repetitive patterns in language models. In *The Fourteenth International Conference on Learning Representations*. ArXiv:2510.15061.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems*, volume 37, pages 68772–68802.

Yihang Peng, Yi Jing, Dewen Liu, Weihan Meng, and Ying Liu. 2026. Not X, but when: Diagnosing a Mandarin LLM writing tic through contextual appropriateness. *arXiv preprint arXiv:2610.22249*.

Mike Perkins, Jasper Roe, Binh H. Vu, Darius Postma, Don Hickerson, James McGaughran, and Huy Q. Khuat. 2024. Simple techniques to bypass GenAI text detectors: implications for inclusive education. *International Journal of Educational Technology in Higher Education*, 21(1):53.

Alex Reinhart, Ben Markey, Michael Laudenbach, Kachatad Pantusen, Ronald Yurko, Gordon Weinberg, and David West Brown. 2025. Do LLMs write like humans? variation in grammatical and rhetorical styles. *Proceedings of the National Academy of Sciences*, 122(8):e2422455122.

Jasper Roe, Mike Perkins, Peter Bannister, Leon Furze, and James Wood. 2026. Dramaturgies of deception: AI humanizers and the performance of legitimacy in higher education assessment. *arXiv preprint arXiv:2605.02649*.

Jenna Russell, Marzena Karpinska, and Mohit Iyyer. 2025. People who frequently use ChatGPT for writing tasks are accurate and robust detectors of AI-generated text. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5342–5373, Vienna, Austria. Association for Computational Linguistics.

Vinu Sankar Sadasivan, Aounon Kumar, Sriram Balasubramanian, Wenxiao Wang, and Soheil Feizi. 2025. Can AI-generated text be reliably detected? stress testing AI text detectors under various attacks. *Transactions on Machine Learning Research*. ArXiv:2303.11156.

Chantal Shaib, Tuhin Chakrabarty, Diego Garcia-Olano, and Byron C. Wallace. 2026. Measuring AI "slop" in text. In *Third Conference on Language Modeling*. ArXiv:2509.19163.

Chantal Shaib, Yanai Elazar, Junyi Jessy Li, and Byron C. Wallace. 2024. Detection and measurement of syntactic templates in generated text. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6416–6431, Miami, Florida, USA. Association for Computational Linguistics.

Zhengyang Shan, Yukyung Lee, and Sophie Hao. 2026. AI writers have a consistent stylometric footprint, but AI editors do not. *arXiv preprint arXiv:2608.27855*. ArXiv comment: EMNLP 2026 main conference; proceedings not yet published at time of access.

SuperAnnotate. 2024. Superannotate/ai-detector. Hugging Face model card, `https://huggingface.co/SuperAnnotate/ai-detector`. RoBERTa-large fine-tune; custom license; last updated 17 Dec 2024; accessed 2026-10-20.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. 2024. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9440–9450, Bangkok, Thailand. Association for Computational Linguistics.

Debora Weber-Wulff, Alla Anohina-Naumeca, Sonja Bjelobaba, Tomáš Foltýnek, Jean Guerrero-Dib, Olumide Popoola, Petr Šigut, and Lorna Waddington. 2023. Testing of detection tools for AI-generated text. *International Journal for Educational Integrity*, 19(1):26.

Wikipedia contributors. 2026. Wikipedia:signs of AI writing. `https://en.wikipedia.org/wiki/Wikipedia:Signs_of_AI_writing`. WikiProject AI Cleanup guide; accessed 2026-10-20.

Yale Undergraduate Admissions. 2026. Policy on the use of AI in application materials. `https://admissions.yale.edu/ai-policy`. Accessed 20 October 2026.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 36, pages 46595–46623.

A Briefs

Set	Genre (n)	Topics
Development	Personal statement (4)	Four fictional students (Portuguese-American shoe-repair family; Nebraska church organist; Kazakh wild-apple collector; a Houston lion-dance performer), Common App prompts
	Short answer (4)	Same students, 250-word supplements (curiosity, roommate note, activity)
	Argument (4)	Phone bans; test retakes; group-project grades; part-time jobs
	Explainer (4)	mRNA vaccines; seasons; photosynthesis; sky colour and sunsets
	Literary analysis (4)	The Great Gatsby; Frankenstein; Macbeth; The Scarlet Letter (four verified quotes each)
Held-out	Personal statement (2)	Two new students: a Polish e-scooter repairer; an Arizona tamale-stand and weather-station student
	Short answer (2)	Why engineering; the world you come from
	Argument (2)	Later school start; graded homework
	Explainer (2)	Noise-cancelling headphones; leap years
	Literary analysis (2)	Romeo and Juliet; Pride and Prejudice
	Email (2)	Extension request to a teacher; cold email to a professor
	Cover letter (1)	Junior counselor at a library coding camp
	Blog post (1)	Robotics team at a regional competition
	Lab discussion (1)	Pendulum period against length
	News article (1)	Reusable cafeteria trays

Table 3: Writing briefs. Every brief names the only permitted source of specifics (interview notes, a facts file, four quotes, or none for arguments and explainers). All people are fictional. The held-out briefs were written after version 6 was fixed and share no example text with any version of the skill.